

Exp No: 16: SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

PROGRAM:

```
#include <reg51.h>

#include <stdio.h>

void serial_ISR(void) interrupt 4
{
    char ch;

    if (RI)
    {
        RI = 0;    // Clear receive interrupt flag

        ch = SBUF;

        // Read received character

        printf("\n>>> INTERRUPT RECEIVED: ");

        printf("%c\n", ch);
    }
}

void delay(void)
{
    unsigned int i;

    for(i = 0; i < 50000; i++); // Delay for visibility
}

void main(void)
{
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled

    TMOD = 0x20; // Timer1 mode 2 (auto-reload)

    TH1 = 0xFD; // Baud rate 9600

    TR1 = 1; // Start Timer1

    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)
```

```

TI = 1;

while(1)

{

// Enable printf

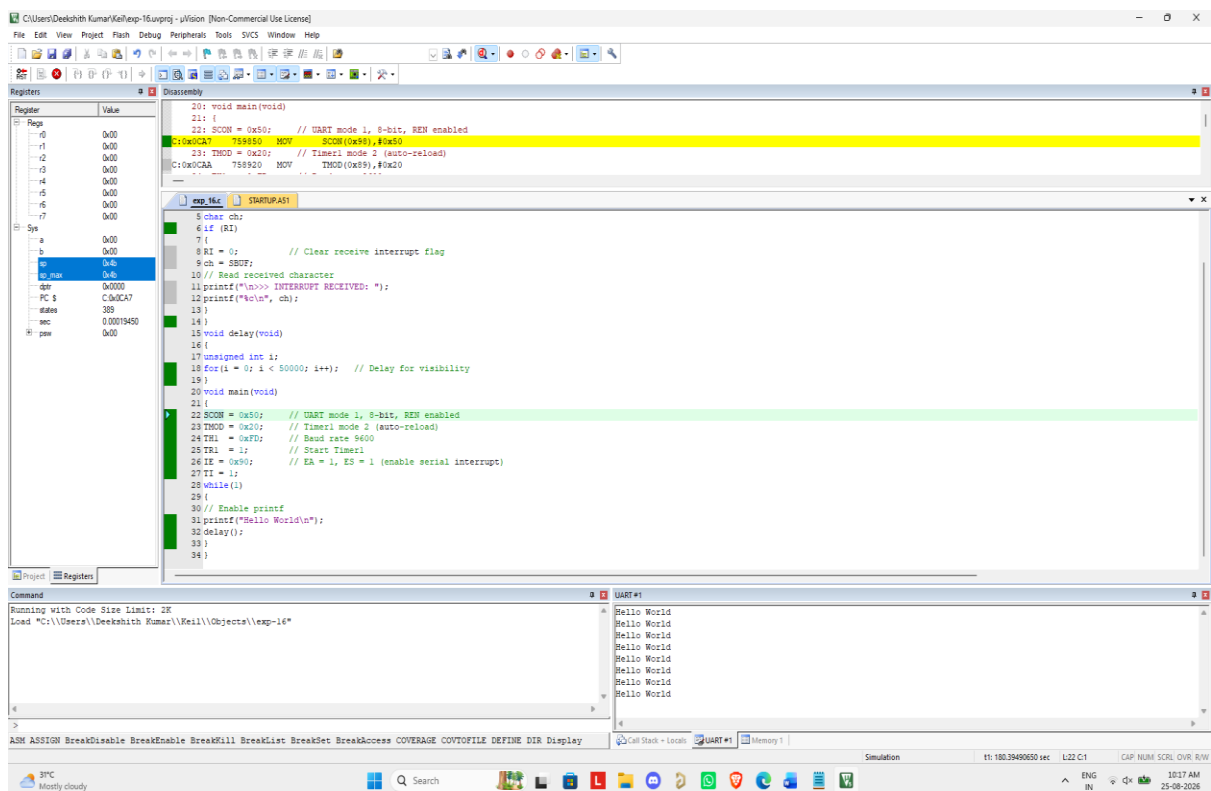
printf("Hello World\n");

delay();

}

```

}Output :



CUUsers\Deekshith Kumar\Keil\exp-16\exp16 - uVision (Non-Commercial Use License)

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Registers

Register	Value
R0	0x00
R1	0x00
R2	0x00
R3	0x00
R4	0x00
R5	0x00
R6	0x00
R7	0x00

Disassembly

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exp16.c

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Command

```
Running with Code Size Limit: 2K
Load C:\Users\Deekshith Kumar\Keil\Objects\exp16
```

UART#1

```
Hello World
Hello World
Hello World
>>> INTERRUPT RECEIVED: a
Hello World
Hello World
Hello World
```

ASH ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE DEFINE DIR Display

Simulation

11:20:43.36161100 sec L:31 C:26 CAP ROMS SCRL OVER 81M

31°C Mostly cloudy

Search

ENG IN 10:58 AM 25-08-2026